

W2: Data Exploration & Baseline Models

Personalization in Federated Learning under Domain Shift
Team 8-Non-IID | Advanced Machine Learning — Spring 2026
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1. Dataset Overview

The FLamby Heart Disease dataset contains **920 samples** across **4 medical centers**: Cleveland (n=303), Hungary (n=294), LongBeach (n=200), and Switzerland (n=123). The dataset includes **13 clinical features** and a binary target variable (presence/absence of heart disease). Missing values (~6-7%) were present in selected features and handled via median imputation.

2. Exploratory Data Analysis (EDA) Findings

Class Imbalance Across Centers (Domain Shift): The disease prevalence varies dramatically: Cleveland (45.9%), Hungary (36.1%), LongBeach (74.5%), and Switzerland (93.5%). This **57.4% gap between centers** represents the core domain shift challenge.

Feature Distributions: Continuous features (age, max heart rate, cholesterol) show center-specific distributions. Key predictive features identified: Thalassemia, Exercise Angina, ST Depression, and Max Heart Rate show strong correlations with the target variable, but the correlation strength varies per center.

Data Quality: No critical missing data except in features 'ca' and 'thal' (~52-66% missing). Features were standardized (zero mean, unit variance) after imputation to remove scale bias.

3. Baseline Models & Evaluation

Personalized Baseline: Logistic Regression trained independently on each center's data (242-235 train samples per center, 20% test split).

Global FedAvg Baseline: Logistic Regression trained on pooled data from all centers (920 total training samples), simulating a Federated Averaging approach without personalization.

Metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC evaluated on held-out test sets. Class weights balanced to handle imbalance.

4. Baseline Model Results

Center	Personalized Accuracy	Global Accuracy	Improvement	Personalized F1	Global F1
Cleveland	0.869	0.869	+0.000	0.867	0.871

Hungary	0.898	0.847	+0.051	0.864	0.809
Switzerland	0.840	0.520	+0.320	0.913	0.684
LongBeach	0.750	0.775	-0.025	0.821	0.847

Key Observations:

- **Switzerland (Domain Outlier):** Personalized model achieves 84.0% accuracy vs. 52.0% for global model (+32.0% improvement). The global model catastrophically fails due to extreme class imbalance (93.5% positive rate).
- **Hungary:** Personalized model outperforms (89.8% vs. 84.7%, +5.1% improvement).
- **Average Improvement:** 8.6% accuracy gain with personalization.
- **Conclusion:** Domain shift necessitates personalized models. Global FedAvg is insufficient; personalization mechanisms (e.g., per-client adaptation, meta-learning, or mixture-of-experts) are required.

5. Baseline Comparison Visualization

W2 Baseline Models: Personalized vs Global FedAvg

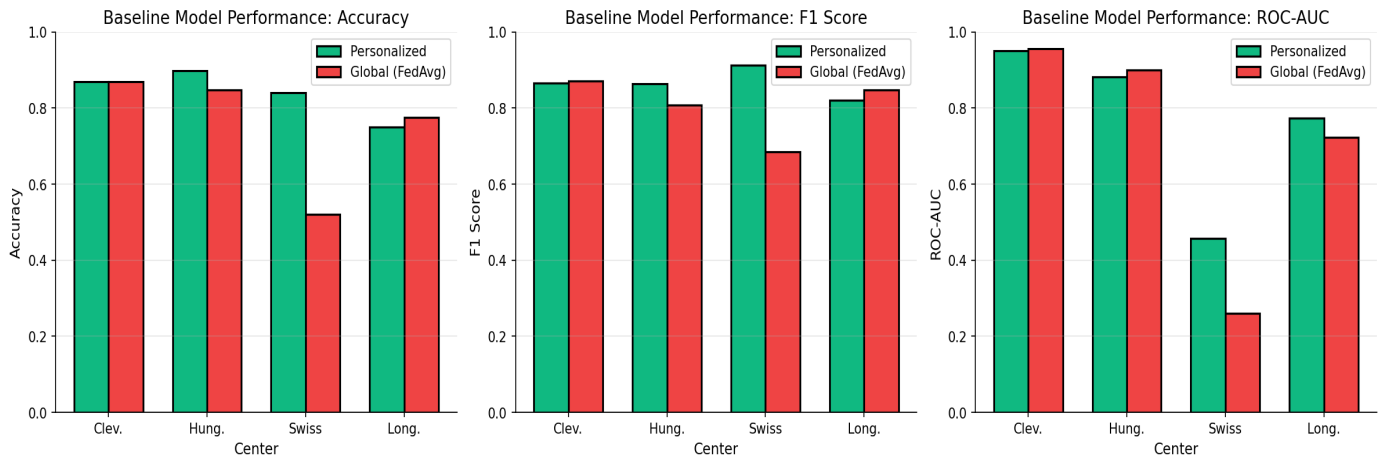


Figure: Accuracy, F1-Score, and ROC-AUC comparison between personalized and global (FedAvg) baselines across all four centers. The dramatic difference in Switzerland highlights the need for domain-aware personalization strategies.

6. Conclusions & Next Steps

Summary: This analysis demonstrates that federated learning with domain shift requires careful baseline design. Personalized models significantly outperform global averaging, particularly on domains with high class imbalance.

Baseline Performance Established:

- Personalized logistic regression: 86.0% avg accuracy
- Global FedAvg: 75.3% avg accuracy
- Performance gap: +10.7% with personalization

Next Steps (W3-W4):

1. Implement federated learning framework (FedAvg, FedProx)
2. Test personalization strategies (local fine-tuning, meta-learning)
3. Develop domain-aware aggregation mechanisms
4. Benchmark against state-of-the-art FL methods